Data Science

TD 3

1. Sampling

Q. 1 Fourier transform of the Dirac comb:

(a) Use the properties of Fourier transform to compute the Fourier transform of

$$\text{Ш}(t) = \sum_{n=-\infty}^{+\infty} \delta(t-n)$$

(b) Compute the Fourier transform the following Dirac comb (we define $\nu_0 = \frac{1}{T}$):

$$\text{Ш}_T(t) = \sum_{n=-\infty}^{+\infty} \delta(t-T\,n)$$

Q. 2 Consider the following signal:

$$x(t) = \sin(2\pi\,\nu_x\,t + \phi)$$

Compute its Fourier transform $\hat{x}$ and plot $|\hat{x}|$:

Q. 3 We defined:

$$y(t) = x(t)\,\text{Ш}_T(t)$$

Compute the Fourier transform $\hat{y}$ and plot y and $|\hat{y}|$.

Q. 4 Redraw the plot when x is a band-limited signal of spectral width B_s. (*i.e.* $\hat{x}(\nu) = 0$ if $|\nu| > \frac{B_s}{2}$)

Q. 5 What will happen if $\nu_e < B_s$?

The sampling of the signal y is performed by defining $y[k] = y(k\,T)$.

2. Reconstructing sampled data

From the samples $y[k]$, we can reconstruct the continuous signal $y(t)$:

$$y(t) = \begin{cases} y[\frac{t}{T}] & \text{if } \frac{t}{T} \in \mathbb{Z}\,, \\ 0 & \text{otherwise.} \end{cases}$$

From $y(t)$, we obtain a reconstruction $z(t)$ of the original signal $x(t)$ by applying a filter with impulse response h:

$$z(t) = (h * y)(t)$$

Q. 6 Knowing that x is a band-limited signal of spectral width B_s, show that the optimal filter that minimizes the reconstruction error $E = \int_{-\infty}^{+\infty} |x(t) - z(t)|^2\,\mathrm{d}t$ is a cardinal sine.

(a) Is this filter causal? What are the issues with such filter?

Q. 7 We design a filter with the following impulse response:

$$g(t) = \frac{1}{\tau} u(t) e^{-\frac{t}{\tau}}$$

(a) Compute the transfer function of this filter g.

(b) Compute its squared modulus $|\hat{g}|^2$ and its phase $\angle(\hat{g})$; plot both quantities:

(c) What is the cut-off frequency ν_c of g:

$$|\hat{g}(\nu_c)|^2 = \frac{1}{2}$$

(d) What could be the time constant τ of this filter to reconstruct x from y?